

SMART CITY TRAFFIC PERFORMANCE REPORT 2026

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Published: January 2026

1. EXECUTIVE SUMMARY

This report evaluates traffic performance across major corridors in the city during the 2025 calendar year. The study focuses on congestion levels, travel times, accident rates, and intelligent transportation system deployments.

Key findings include:

- Average city-wide traffic volume increased by 12%.
- Peak-hour congestion increased by 8%.
- Travel times increased by 6% in central business districts.
- AI-based adaptive traffic signal systems reduced delays by up to 18%.

2. TRAFFIC VOLUME ANALYSIS

Daily traffic counts were collected using roadside sensors and camera systems.

Table 1: Average Daily Vehicle Counts

Central Avenue: 48,500 vehicles/day

Metro Avenue: 52,200 vehicles/day

Ring Road: 64,800 vehicles/day

Airport Corridor: 38,900 vehicles/day

Morning Peak Period:

07:00 AM – 09:00 AM

Evening Peak Period:

05:00 PM – 07:00 PM

The highest peak-hour volume was recorded on Ring Road with approximately 6,400 vehicles per hour.

3. CONGESTION HOTSPOTS

The following intersections experienced severe congestion:

A. Metro Ave & 5th Street

Level of Service: E

Average Vehicle Delay:

95 seconds per vehicle

Queue Length:

420 meters during evening peak

Primary Cause:

Commercial development and increased pedestrian crossings.

B. Central Ave & Park Junction

Level of Service: D

Average Vehicle Delay:

62 seconds per vehicle

Queue Length:

280 meters

Primary Cause:

High bus volumes and frequent curbside parking activity.

4. ROAD SAFETY ANALYSIS

A total of 2,184 traffic incidents were reported during 2025.

Severity Breakdown:

Minor Incidents:

1,624

Major Incidents:

486

Fatal Incidents:

74

The majority of incidents occurred between 06:00 PM and 09:00 PM.

Top Accident Locations:

1. Ring Road Exit 7
2. Metro Ave & 5th Street
3. Airport Corridor Interchange

5. INTELLIGENT TRANSPORTATION SYSTEMS

Several smart transportation initiatives were deployed.

Adaptive Traffic Signal Control (ATSC)

Deployment Locations:

- Metro Avenue
- Central Avenue
- Ring Road

Results:

Average Delay Reduction:

18%

Average Travel Time Reduction:

11%

Fuel Consumption Reduction:

7%

The ATSC system uses AI-based prediction models to dynamically adjust signal timing based on real-time traffic demand.

6. PUBLIC TRANSPORT PERFORMANCE

Bus Operations:

Average Daily Ridership:

412,000 passengers

Average Bus Speed:

24 km/h

On-Time Performance:

87%

Metro Operations:

Average Daily Ridership:

610,000 passengers

On-Time Performance:

94%

7. RECOMMENDATIONS

1. Expand adaptive signal control to all major corridors.
2. Introduce AI-powered congestion prediction systems.
3. Upgrade roadside camera infrastructure.
4. Improve pedestrian crossing management at high-volume intersections.
5. Increase public transport capacity during peak periods.

8. CONCLUSION

The city continues to experience growing transportation demand. Intelligent transportation technologies have demonstrated measurable improvements in traffic flow and safety.

Continued investment in AI-driven traffic management is recommended to support future mobility needs.